Vinamra Gupta

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EDUCATION

Indian Institute of Technology, Jodhpur — **Aug 2024 - Present**
Bachelor of Technology - Artificial Intelligence and Data Science - CGPA - 7.92/10.0 — Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, React, Vite, HTML, CSS, Data Structures, Graph Algorithms
Machine Learning & Libraries: PyTorch, Pandas, Numpy, LLM fine-tuning (LoRA, QLoRA), TensorFlow, scikit-learn
Tools: Git, GitHub, VS Code, Ubuntu, Jupyter Notebook, Colab
Areas of Interest: Competitive Programming, Deep Learning, Financial Machine Learning

EXPERIENCE

Undergraduate Researcher — **Jun 2025 - Present**
Dep. of CSE, IITJ (Supervisor: Dr. Angshuman Paul) — Jodhpur, Rajasthan

- **Conceived** a **wavelet-domain MRI denoising framework** using **Rician noise simulation**, expediting restoration of **high-NSA MRI images** from fast, **low-NSA acquisitions**, reducing simulated scan times by up to **75%**.
- **Evaluated 6** deep learning paradigms including **CNNs, AEs, VAEs, Diffusion Models, and Restormer variants** (DWT/SWT-based); yielded **32.53 PSNR, 0.82 SSIM, 0.05 LPIPS, and 17.12 FID** on **1 to 8 NSA** recovery with **60% recovery** on downstream segmentation tasks, outperforming **5 current SOTA models**.

Project Member (Inter IIT Tech Meet 14.0 - 7th Place) — **Oct 2025 - Dec 2025**
Ebullient Securities Challenge (High Preparation Problem Statement) — Jodhpur, Rajasthan

- **Formulated** a **quantitative trading strategy** on mid-frequency, **anonymized multi-feature time-series data** under market constraints, processing **over 200 masked features** spanning price, volatility, volume and bands.
- **Streamlined** multiple **adaptive indicators and regime-switching models**, including **volatility-based KAMA** and **Kalman filters**; generated **21% and 32% annualized returns** delivering **Calmar ratios of 5.33 and 14.40** with **max drawdowns restricted to 3.83% and 2.18%** on blind out-of-sample data.

PROJECTS

Food Image Classification | GitHub | **scikit-learn, PCA, Python** — **Feb 2026 - Apr 2026**

- **Accelerated** a **4-phase classical ML pipeline** for **20-class food image classification**, integrating **4 handcrafted features (HOG, LBP, GLCM, Histogram)** compressed by **80% via PCA**, alongside a custom **Bayesian KDE** and **7 GridSearchCV-tuned classifiers** evaluated via **stratified 5-fold cross-validation**.

Trading Strategy | GitHub | **Python, Statistical Finance, Backtesting** — **Oct 2025 - Dec 2025**

- **Architected** a **Python-driven backtesting pipeline** evaluating **>500,000 data points**, integrating **volatility-adjusted position-sizing** and **automated risk controls** to navigate masked price-volume behaviors and **cap maximum drawdown at 2-3%** even during **severe market downturns**.

CashFlow Minimizer-Visualizer | GitHub | Deployed Link | **C++, Graph Algorithms** — **Sep 2025 - Nov 2025**

- **Produced** a **transaction optimization platform**, trims redundant cash flows by **90%** using **graph algorithms**, featuring a **real-time Web Worker-driven dashboard** to benchmark **5 algorithms (Greedy, Min Cash Flow, Heap, Priority Queue, and Sorting-based)** on execution latency and efficiency with interactive visualizations.

KEY COURSES TAKEN

- Data Structures & Algorithms
- Operating Systems
- Computer Networks
- Computer Architecture
- Machine Learning
- Linear Algebra
- Probability & Statistics
- Intro to Computer Science

ACHIEVEMENTS

- Specialist at Codeforces (Max Rating: 1544)
- 7th rank among 23 IITs in Inter IIT Tech Meet 14.0 (Ebullient Securities (Quant PS))
- JEE Advanced 2024, Secured AIR 3048

CAMPUS INVOLVEMENT & LEADERSHIP

- **Core Member**, Quant Club, Indian Institute of Technology, Jodhpur — **Aug 2025 - Present**
- **Assistant Head**, Prometeo'26, Indian Institute of Technology, Jodhpur — **Nov 2025 - Jan 2026**